Birla Institute of Technology & Science, Pilani
Pilani Campus
AUGS/AGSR Division

GPIC 2026 | Gujarat Police Innovation Challenge

Module 01: PRAHARI

Lectures Hackathon demo · Amit Dua, Arnav, Aria · September 2026

1.1 Official problem

The Home Department of Gujarat published the challenge at `https://sentinel.gujarat.gov.in/problems`. A word-for-word transcript lives in `06_References/SENTINEL_Problems_Page.md`, crawled 03 September 2026. Phase 1 submission closes 07 September 2026 at 12:00 IST. PRAHARI is Category 1 (student). The team is Amit Dua, Arnav, and Aria.

Twenty-six departments run independent CCTV estates. Cameras are analog and IP. Sites run from border districts to Valsad, Dahod, Somnath, Jamnagar, and Dwarka. Some NVRs keep seven days, some fifteen. Watchlists already exist: VAHAN, SARTHI, eGujCop, AFIS, NAFIS. The cameras do not match against those lists. There is no statewide census of which public-domain cameras exist, whether they are alive, or where coverage stops.

The official core goal is a secure, scalable, interoperable, technically feasible, and cost-effective approach that uses existing infrastructure to the maximum practical extent. Replacing every departmental VMS in a student PoC is not feasible. A central recorder that ingests 80,000 full-HD streams for 15 days needs statewide dark fibre, petabytes of hot storage, and a GPU farm that this team cannot claim. PRAHARI therefore adds an intelligence plane on top of what already records video.

1.2 Hybrid model

The official page lists four reference models and permits a hybrid. Model 1 is the registry and GIS foundation and may support Models 2, 3, and 4. Model 2 connects directly to departmental streams through RTSP, ONVIF, or vendor APIs and may run ANPR without storing every frame. Model 3 inserts a federation middleware. Model 4 is a full central VMS with recording, tiered storage, and statewide AI.

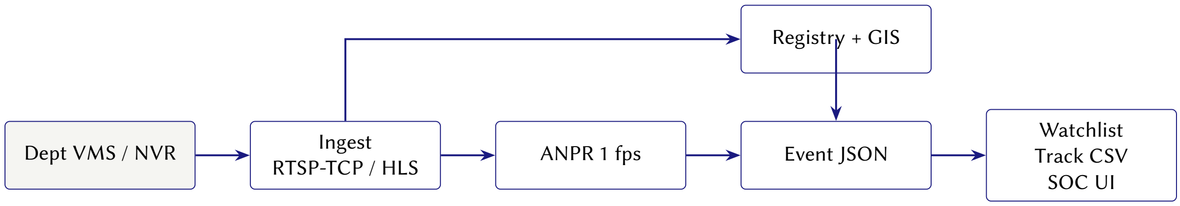
Definition 1.2.1 (PRAHARI hybrid)**.** Model 1 supplies camera metadata, GIS, health, and gap reports. Model 2 opens reachable RTSP, HLS, WHEP, or ONVIF streams and runs ANPR on sampled frames. Thin Model 3 is a detection-event JSON bus, an in-memory watchlist matcher, and `GET /api/track/{plate}`. Model 4 central recording is written as Phase-2 for selected cameras only.

Key idea

> A new vendor is one adapter. The detection JSON does not change if the bus later becomes Kafka.

The official architecture principles ask for an open, modular, standards-based, vendor-neutral design. PRAHARI keeps departmental VMS in place. Adapters speak RTSP-TCP, HLS, WHEP, and ONVIF. Kafka, Kubernetes, and Ceph are named as later swap points. They are not in this PoC.

1.3 Architecture



The PoC runs FastAPI and SQLite on one laptop. The UI is Leaflet plus vanilla JS on port 8080. Capture uses OpenCV with FFmpeg. ANPR uses morphology plus Tesseract behind recognize(). Alerts move on an in-process WebSocket.

PoC versus later swap

Piece	PoC	Later swap
API	FastAPI + Uvicorn	same HTTP contract
UI	Leaflet + vanilla JS	React if a wall needs it
Store	SQLite	PostgreSQL + PostGIS
Capture	OpenCV + FFmpeg, RTSP TCP	DeepStream on regional GPUs
ANPR	morphology + Tesseract	YOLO plate detector + PaddleOCR
Alerts	in-process WebSocket	Redis / Kafka
Auth	HTTP Basic + signed cookie	department SSO

1.4 Live Sentinel contract

The sandbox publishes each camera as a live RTP/RTSP stream. On 01 September 2026 the host live.corp8.cloud answered HTTP 502 on /api/ingest. On 03 September 2026 the live portal is https://cctv.corp8.cloud/. Catalogue GET /cameras.json, after POST /auth/login, returns a JSON array of {id, name}. Thirty cameras were listed (cam01 to cam30). /api/ingest returns HTTP 404 on this host.

HLS playlists live at https://cctv.corp8.cloud/<id>/index.m3u8. The origin requires the access cookie and a browser User-Agent. A plain curl without those headers returns the text browser required. RTSP is not on the TLS hostname. The live /resource page publishes rtsp://103.250.160.189:8554/stream/<id>. Port 8554 on that address is open from the team network.

The capture client sets OPENCV_FFMPEG_CAPTURE_OPTIONS=rtsp_transport;tcp before it imports OpenCV. Event time is presentation timestamp (CAP_PROP_POS_MSEC). Packet arrival time and declared frame rate are unused. Reconnect backoff starts at 2 s and caps at 30 s. Decoder warnings at join (Error constructing the frame RPS) are logged and ignored until the first IDR. A PTS rewind or a gap above 5 s is treated as a scene cut because each feed loops.

Warning

> There is no file download. /stream/<id> answers range requests for a player. curl or wget of that path yields a partial file that looks complete. The client consumes live. It does not publish to the gateway. OpenCV may print SETUP failed: 461 Unsupported Transport once (UDP). TCP then succeeds. That line is not a down camera.

Implementation

> app/services/catalogue.py logs in with SENTINEL_PASSWORD, reads /cameras.json, and fills HLS and RTSP only when the JSON omitted those URLs. Camera ids still come from the JSON. The browser never receives the Sentinel cookie. Stream tiles use a 60-second HMAC token on /api/stream/{id}.

1.5 Measured government-feed result

Numbers in this section are MEASURED on 03 September 2026 unless labelled otherwise.

The registry upserted all 30 catalogue cameras with department=Sentinel and cam_type=sandbox. Sequential RTSP-TCP probes of cam04 (Paldi Circle) and cam01 (Chimanbhai Bridge) returned health live. One frame from cam04 was written to 03_Data/recordings/first_live_frame.png: 1920×1080, PTS 1080 ms.

The live manifest has no live flag and no coordinates. The Operations map therefore hides those rows (it already skips lat $= 0$, lon $= 0$). The Cameras table lists them and offers Open tile. Gap reports flag missing coordinates honestly.

Tesseract is not on PATH on the build laptop. recognize() raises TesseractNotFoundError. Operator confirm (POST /api/ingest/confirm) wrote plate GJ01AB1234 on camera cam04 with confidence 1.0. The matcher raised a CRITICAL alert. The script gov_report.py wrote one catalogue row to gov_feed_plates.csv with the note “OCR empty, confirm used”.

1.6 ANPR and the detection event

Analytics samples one frame per second using PTS deltas, not sleep(1) and not declared FPS. The plate pipeline is: localise a plate-like box, run Tesseract with an alphanumeric whitelist, then normalise.

Definition 1.6.1 (Indian plate normaliser)**.** After stripping spaces and hyphens, the string must match ^[A-Z]{2}[0-9]{1,2}[A-Z]{1,3}[0-9]{3,4}$. GJ 01 AB 1234 and GJ-01-AB-1234 both become GJ01AB1234. Anything else is dropped.

Implementation

> recognize(frame_bgr) returns {plate, plate_raw, confidence, crop}. Callers never import Tesseract. A YOLO engine can replace the body when ANPR_ENGINE=yolo. Operator confirm writes confidence 1.0 and still hits the matcher, so a readable plate cannot fail a live demo if OCR misses.

Frozen event fields: event_id, plate, plate_raw, confidence, camera_id, lat, lon, ts, pts_ms, crop_uri, category, priority, source_case_id.

1.7 Watchlist, track, security

The matcher keeps plates in a Python `set`. Lookup is $O(1)$. The set reloads from SQLite every 60 s. Duplicate suppression: the same plate on the same camera within 120 s increments `counter` on the open alert. Priority CRITICAL is stolen or wanted. `WL-001` is `GJ01AB1234` / STOLEN and cannot be deleted.

Designated vehicle `GJ01AB1234`

> `GET /api/track/GJ01AB1234` returns the six seed points in this order: CAM-VAL-001 Valsad 06:12, CAM-SUR-001 Surat 07:48, CAM-AHD-004 Narol–Naroda 09:05, CAM-AHD-001 SG Highway 09:22, CAM-GNR-003 Koba 09:51, CAM-GNR-001 Gandhinagar Sector 21 10:04. On 03 September 2026 a live confirm on `cam04` appended. The detection table then held eight cameras for that plate. The six seed ids remained. CSV columns are plate, camera_id, ts, confidence, category on the government report, and the longer GIS columns on `/api/track/{plate}/report.csv`.

Roles: `superadmin`, `soc_operator` (`judge`), `dept_viewer` (`home.viewer` sees Home only), `auditor` (GET only). Stream tiles use a 60-second HMAC token. Responses to the browser omit `rtsp://`. Private-permitted cameras require `consent=true`. Audit rows cover onboard, watchlist edit, ack, report download, and confirm. Stream URLs are not stored in audit detail.

1.8 Official evaluation, mapped

The problems page scores seven common areas. Bonus does not rescue a failed mandatory item.

1. Successful test case on the government feed. MEASURED: 30 cameras onboarded from the live catalogue; RTSP-TCP frame on disk; confirm and CSV on `cam04`. Screen-record of that path is still a human task.
2. Solution presentation. This notes file and `slides.tex` (BITS Beamer, logo top right) cover model justification, architecture, analytics, watchlist, stack, scale, and impact.
3. Solution architecture. HLD at `04_Documents/PRAHARI_HLD.md` matches the running APIs. Model 4 is not faked.
4. Working platform. Own-feed file `own_feed.mp4` and the live catalogue both write detection rows. Mock UIs without a database row are rejected by the official page and by this team.
5. Video analytics. Confirm path is MEASURED. OCR is blocked until Tesseract is on PATH. Face recognition is not the demo.
6. Scalability and PoC readiness. Section 1.9 is DESIGN TARGET. The laptop count is MEASURED.
7. Submission completeness. GitHub `https://github.com/amitduabits/PRAHARI` is live. Unlisted YouTube, Drive Anyone+Viewer, and a hosted URL remain human work before 07 September 2026.

Bonus already in the product, and scoring only if a video or this deck shows it: hybrid justification, cross-camera track, RBAC and audit, gap report (Dahod offline, short retention), WebSocket alerts, 1 fps regional cost sentence.

1.9 Scale and cost

Numbers below are DESIGN TARGET for a statewide intelligence plane, not measured laptop throughput.

Assume 45,000 public-domain cameras run ANPR at 1 fps with an 80 KB JPEG crop. Naive centralisation is $45{,}000 \times 80\,\mathrm{KB/s} \approx 3.6\,\mathrm{GB/s}$. Five regional GPU sites (Ahmedabad, Surat, Rajkot, Vadodara, Bhuj) cut that to about 720 MB/s per site. The SOC wall pulls 16 to 64 HLS tiles. Everything else is metadata and crops. Low-connectivity outposts send events only.

Order-of-magnitude yearly cost of that plane, not of replacing 26 VMS estates: GPU nodes 1.8–2.4 Cr, object storage 0.8–1.2 Cr, bandwidth 0.6–1.0 Cr, twelve engineers 1.5 Cr; total about 5–6 Cr.

Hot / warm / cold recording of every frame is Model 4 and Phase-2, selected cameras only.

1.10 How to run the PoC

Clone `https://github.com/amitduabits/PRAHARI`. From `02_Code/prahari` copy `.env.example` to `.env`. Set the judge password before any hosted demo. Set the Sentinel host, password, and RTSP IP after the live portal login. Run `run.ps1` (Windows) or `run.sh` (Unix). Open `http://127.0.0.1:8080` and sign in as `judge`. Tests on 03 September 2026: 33 passed, 1 skipped (Tesseract binary). Integrator checks live in `test_integrator_laws.py`.

Three-minute walk: Operations map (seeded pins, Dahod offline), Cameras table Open tile on `cam04`, Vehicle Track reconstruct `GJ01AB1234` and download CSV, Alerts CRITICAL and Ack, Onboard confirm if OCR misses, Analytics & Gaps for short-retention Food and Civil Supplies cameras.